

Business Analytics Report

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1. Dataset Overview

Total Rows: 125751

Total Columns: 16

Missing Values: 0

2. Column Information

- Transaction Date: object
- Sales Type: object
- Is Return: int64
- Supplier: object
- Product No: object
- Product Description: object
- Product Division: object
- Product Category: object
- Product Subcategory: object
- Product Segment: object
- Store: object
- Sales Channel: object
- Qty Sold: float64
- Sales Amount: float64
- Cogs: float64

3. SQL Analysis Results

Query 1 - Top Categories:

- Rank 1: Full Price
- Rank 2: Promo
- Rank 3: Markdown Tier 1
- Rank 4: Markdown Tier 2
- Rank 5: Clearance

4. Statistical Metrics

- Total Rows: 125751
- Total Columns: 16
- Missing Values: 0

- Duplicate Rows: 0
- Is Return - Total: 7491
- Is Return - Average: 0.06
- Is Return - Median: 0.0
- Is Return - Std Dev: 0.24
- Qty Sold - Total: 116995.0
- Qty Sold - Average: 0.93
- Qty Sold - Median: 1.0
- Qty Sold - Std Dev: 0.59
- Sales Amount - Total: 10482110.25
- Sales Amount - Average: 83.36
- Sales Amount - Median: 90.0
- Sales Amount - Std Dev: 83.66
- Cogs - Total: 5847751.1
- Cogs - Average: 46.5
- Cogs - Median: 50.05
- Cogs - Std Dev: 43.6

5. AI Analysis & Recommendations

Business Domain:

This dataset appears to be from an e-commerce or retail business. The data includes information about transactions, such as date, sales type, product details, and financial metrics, suggesting that it's a dataset used for sales analysis, inventory management, or supply chain optimization.

Key Findings:

1. High Return Rate: The average return rate is 0.06, suggesting that nearly 1 in 17 transactions results in a return. This could indicate issues with product quality, customer satisfaction, or logistics.
2. Low Average Sales Amount: The average sales amount is \$83.36, which is relatively low. This might suggest a business model focused on high-volume, low-value sales or a need to optimize pricing strategies.
3. High Standard Deviation in Sales Amount: The standard deviation of sales amount is \$83.66, indicating a significant spread in transaction values. This might be due to variations in product prices, sales channels, or customer segments.

Business Risks:

1. High Return Rate: The high return rate could lead to increased costs, damaged customer relationships, and reduced profitability.
2. Inventory Management Challenges: The dataset's focus on transactions and financial metrics suggests that inventory management might be a challenge. Insufficient inventory, overstocking, or inventory obsolescence could negatively impact the business.
3. Unoptimized Pricing Strategies: The low average sales amount and high standard deviation in sales amount might indicate that pricing strategies are not optimized for the business, resulting in lost revenue opportunities.

Recommendations:

1. **Analyze Return Rate Drivers:** Investigate the causes of the high return rate, such as product quality issues, customer dissatisfaction, or logistics problems. Identify areas for improvement to reduce returns and minimize costs.
2. **Optimize Pricing Strategies:** Conduct a thorough analysis of pricing strategies to determine potential opportunities for revenue growth. Consider factors like product segments, sales channels, customer segments, and competitor pricing.
3. **Review Inventory Management Processes:** Evaluate inventory management processes to ensure they are optimized for the business. This might involve implementing inventory management software, conducting regular inventory audits, or adjusting supply chain logistics.

6. Forecast

- May 2026: 11,116.16
- Jun 2026: 10,982.19
- Jul 2026: 10,848.22
- Aug 2026: 10,714.25
- Sep 2026: 10,580.27
- Oct 2026: 10,446.30
- Nov 2026: 10,312.33
- Dec 2026: 10,178.35
- Jan 2027: 10,044.38
- Feb 2027: 9,910.41
- Mar 2027: 9,776.44
- Apr 2027: 9,642.46